

LANGCHAIN CORE CONCEPTS CHEATSHEET

AI Engineer Ready · Week 3 — The Building Blocks Before You Write Code

Golden Rule → prompt / messages → chat model → response → (optional) parse → use in next step

1. What Is LangChain?

LangChain is a **framework** for building LLM apps step by step. Think of it as a toolbox that helps you organize prompts, call models in a consistent way regardless of provider, connect multiple steps together (chains), and reuse the same logic across projects. You don't *need* LangChain to call an LLM — you could call the OpenAI API directly. It earns its keep once you have more than one step, or want to swap models without rewriting your app.

2. LLM vs Chat Model

	LLM	CHAT MODEL
Input	Plain text prompt	List of role-based messages
Output	Text completion	A message object (<code>.content</code>)
Roles	None	system, user, assistant
Best for	Simple text completion	Conversation, structured workflows
Used in this course	No	Yes

3. Message Roles

```
from langchain.messages import SystemMessage, HumanMessage

messages = [
    SystemMessage(content="You are a helpful tutor."), # sets behavior
    HumanMessage(content="Explain LangChain in one sentence."), # the ask
]
```

ROLE	PURPOSE
system	Hidden instruction — sets tone, persona, rules. The user never sees this directly.
user (HumanMessage)	What the person actually typed or asked.
assistant (AIMessage)	The model's previous replies, if keeping conversation history.

Rule of Thumb

Put *stable* instructions in the system message. Put *variable* input in the user message.

4. Must-Know Glossary

TERM	MEANING
Prompt	The instruction you send to the model
PromptTemplate	A reusable prompt with <code>{placeholders}</code> filled in at runtime

TERM	MEANING
Chain	One step's output feeding into the next step's input
Token	A chunk of text (~4 characters) the model reads/writes
Context window	Max tokens the model can process at once (gpt-4o-mini ≈ 128k)
Temperature	0–1 knob for randomness (0 = deterministic, 1 = creative)
invoke / batch / stream	The three ways to call a chat model — see Cheatsheet 3